

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
3. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
4. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.
5. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.
6. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
7. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
8. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
9. Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.
10. An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary detection approach, describing its working and effectiveness.

96
100

Code of Conduct:

Haritha (19242525) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**Rubrics for Evaluation**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Image appears too bright

Suitable grey level transformation: negative transformation

Concept

A negative transformation converts bright pixels into dark pixels into bright pixels.

Application process:

- * Read the image
- * Apply the formula

$$S = L - 1 - r$$

Where:

- r = input pixel value
- L = maximum intensity (256 for 8-bit image)
- display the transformed image.

Justification

it reduces excessive brightness and makes hidden details more visible

Expected outcome

- * Bright regions become darker
 - * Image visibility improves
 - * fine details become clearer.
- it reduces excessive brightness and makes hidden details.

2, Noisy image with Random intensity variations

Suitable Smoothing filter

medium filter

Concept

A median filter removes random noise by replacing the center pixel with the median value of its neighbouring pixels.

Working:

- * Select a Small window (3x3)
- * Arrange pixel values in ascending order
- * Replace the center pixel with the median value

Justification

It removes Salt and pepper noise while preserving image edges.

Impact:

- Noise is reduced
- Edges remain sharp
- image becomes smoother.

3, Image appears blurred after smoothing

Suitable Sharpening method: Laplacian Sharpening

Concept:

The laplacian operator highlights rapid intensity changes (edges)

Working

→ Appl

Ju

Working

- Apply the laplacian filter
- Add the result back to the original image

Justification

it reduces restores lost details after smoothing

Expected outcome

- Edges before Sharpen
- image looks cleaner
- fine details are restored.

4) fine details are not visible clearly.

Suitable filtering Technique: High-pass filter

Concept

A high pass filter enhances high frequency components such as edges and fine details

Working

- Working removes low-frequency (smooth) information
- preserves high-frequency information

Justification

fine details are represented by high frequencies

Effect

- Image become Sharper
- Small details become visible.
- Edge Contrast increases.

5, Noisy Image Before Analysis:

Suitable spatial filtering method: Gaussian filter

Concept:

A Gaussian filter smooths an image using weighted averaging

Working

- Apply a gaussian kernel (3x3 or 5x5)
- neighbouring pixels contribute based on the distance.

Justification

it reduces gaussian noise before image processing

Result:

- noise decreases
- image becomes smoother
- Suitable for further analysis

6, Object is brighter than background

Suitable Segmentation technique: Threshold Segmentation

Concept

Thresholding separates objects based on pixel intensity.

Working:

- * Select threshold value T
- * if $\text{pixel} > T \rightarrow \text{object}$
- * Otherwise $\rightarrow \text{background}$

Justification
Works well
back

Justification

Works well when the object is much better than the background.

Effectiveness:

- fast
- Easy to implementation
- produces clean object separation.

7. Separate foreground and background

Suitable method: global thresholding

Concept:

One threshold value is applied to the entire image

Process

- choose threshold T
- pixels above T become white
- pixels below T become black

Justification:

Simple and effective when illumination is uniform.

Result:

- foreground is separated from the background
- Binary image is obtained.
- Simple and effective when illumination is uniform.

8) Identify object Edges

Suitable Edge detection Method: Canny Edge detector

Concept:

Canny detects Strong and Continuous edges with less noise.

Working:

1. Smooth image
2. Calculate gradients
3. Apply non-maximum Suppression
4. use double threshold
5. perform edge tracking

Justification

produces accurate and thin edges

Effectiveness:

- detects true edges
- Reduces false edges
- gives high accuracy

9, Noise fragmentation edges:

Suitable method: morphological closing

Concept

closing fills small gaps and connects the broken edges.

Edge detector

Working:

- perform dilation
- Then perform erosion

Justification

Improves edge continuity

Role in Segmentation:

- Connects broken boundaries
- removes small holes
- produces smooth object outlines

10) Precise object boundary extraction

Suitable approach: Canny edge detection + contour

Concept:

Canny finds accurate edges, while contour detection extracts complete object boundaries.

Working

- detect edges using canny
- find contours around connected edges
- draw object boundaries.

Justification:

Provides accurate object shapes for analysis.

Effectiveness:

- precise boundary extraction

→ Accurate shape analysis

→ Suitable for object recognition and measurement